SRS 1

1. INTRODUCTION

1.1 Purpose and Overview

Threads of Life is an interactive, narrative-driven game exploring moral reflection through player choices across cycles of life, death, and reincarnation. Unlike conventional games focused on winning or scoring, it immerses players in emotionally charged scenarios where actions have lasting ethical consequences.

The project merges narrative design and AI-driven sentiment analysis to interpret player input and dynamically adapt storylines. It positions games as digital tools for introspection, empathy, and moral reasoning, combining creative storytelling with technical innovation.

1.2 Problem Statement

Modern games often emphasize entertainment and competition, rewarding efficiency over empathy. Moral systems—when present—tend to reduce complex ethical dilemmas into binary metrics like “good” or “evil.” This simplification diminishes moral engagement and limits opportunities for players to reflect on their decisions.

There is a need for interactive experiences that foster awareness of consequence, empathy, and self-reflection, especially for younger audiences who spend significant time in digital environments.

1.3 Proposed Solution

Threads of Life presents morally ambiguous scenarios where players express themselves through natural language input rather than pre-set options. A sentiment analysis module interprets emotional tone (e.g., empathetic, neutral, hostile) and maps it to narrative consequences using Unity and Ink scripting.

Each decision affects relationships, future scenarios, and reincarnated “lives,” producing a tapestry of interconnected outcomes.

There are at least six narrative scenes that explore key moral themes:

1. **Witnessing an Accident** – Responsibility as a bystander

2. **Bus Dilemmas** – Empathy and honesty in daily life

3. **Workplace Abandonment** – Integrity under pressure

4. **Credit Theft** – Moral courage in hierarchies

5. **Restaurant Injustice** – Compassion without reward

6. **The Garden of Reflection** – Forgiveness and renewal

1.4 Scope

The system comprises five major modules:

- **Narrative Engine**: Controls story branching, state persistence, and level transitions using Unity + Ink.

- **Sentiment Analysis Module**: Processes typed player responses using pretrained NLP models or APIs.

- **Dialogue & Consequence Manager**: Maps emotional classifications to story outcomes and character reactions.

- **Reflection Journal**: Summarizes moral choices and personal growth after each level.

- **Data Logging System**: Records all decisions for feedback and analytics.

1.5 Intended Users

- **Primary Users:** Players (ages 12–30) interested in story-driven, reflective gameplay.

- **Secondary Users:** Educators and researchers exploring moral cognition, ethics, or human–computer interaction.

2. FUNCTIONAL REQUIREMENTS

The following sections define the system's core functions, organized into five modules.

2.1 Module 1: Narrative Engine

Purpose: Manage story progression, world-state, and branching logic.
Functions:

- **Story Progression Control:** Load current scenes and transition based on prior choices.
- **Choice & Consequence Management:** Update moral variables, track cumulative outcomes.
- **Reincarnation Mechanic:** Carry moral data across levels to simulate karmic continuity.
- **Level-Specific Management:** Handle unique dilemmas, NPC reactions, and alternate endings per scenario.

2.2 Module 2: Player Interaction System

Purpose: Enable natural, expressive communication with the game.
Functions:

- **Text Input Interface:** Capture and validate player text responses at key decision points.
- **Interactive Dialogue Display:** Present adaptive dialogue based on sentiment output.
- **Hybrid Choice System:** Combine structured options with open-ended typing for accessibility and expressiveness.

2.3 Module 3: Sentiment Analysis Module

Purpose: Interpret player text and classify it into emotional categories.
Functions:

- **Input Preprocessing:** Tokenize, normalize, and clean text.
- **Sentiment Classification:** Use pretrained NLP models (e.g., Hugging Face) to detect tone (positive, neutral, negative; empathetic, hostile, regretful, etc.).
- **Narrative Mapping:** Match sentiment scores to corresponding Ink branches.
- **Fallback Handling:** Offer clarification or default neutral paths when confidence is low.

2.4 Module 4: Reflection and Feedback System

Purpose: Reinforce self-reflection and awareness of moral consequence.
Functions:

- **Reflection Journal:** Summarize player actions, what they could have done better.
- **Moral Visualization:** Represent outcomes visually (e.g. garden bloom).
- **Post-Game Feedback:** Provide narrative summaries.
- **Level Endings:** Link each level's outcome to the next life's starting condition.

2.5 Module 5: System Infrastructure & Analytics

Purpose: Ensure stability, scalability, and research data collection.
Functions:

- **Data Management:** Maintain a secure local database of decisions and states.
- **Analytics:** Generate anonymized metrics for research
- **Error Handling:** Implement crash logging and sentiment fallback.

2.6 USE CASE DIAGRAMS

Figure 1: Core Gameplay Flow

Primary user interactions and game progression

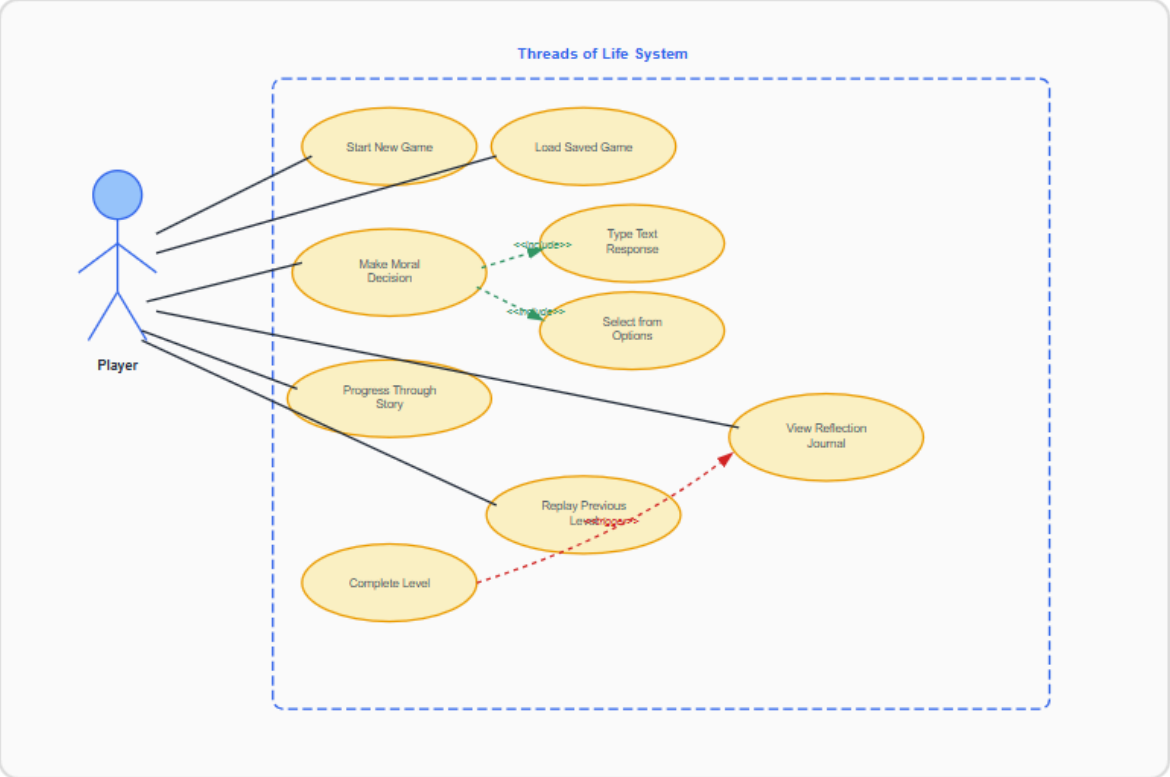
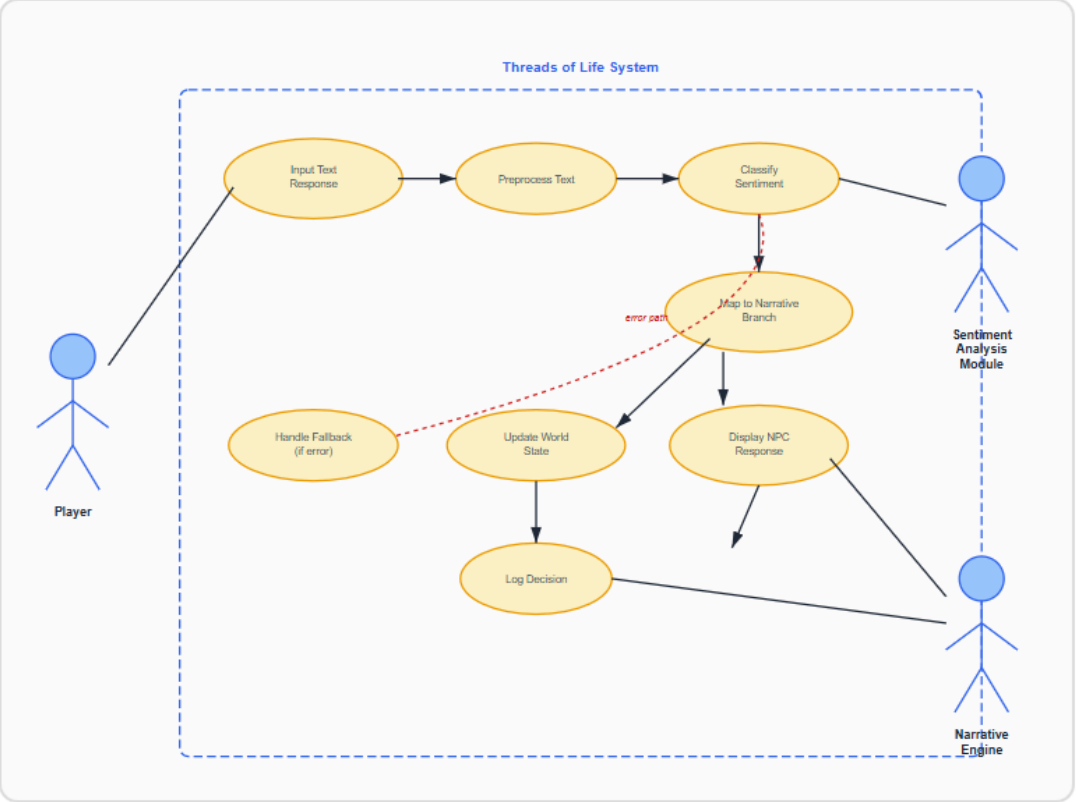


Figure 2: Sentiment Analysis & Narrative Branching

System processing flow for text-based moral decisions



3. NON-FUNCTIONAL REQUIREMENTS

3.1 Performance

- **Response Time:** Sentiment classification + narrative output < 2s.
- **Frame Rate:** ≥ 60 FPS on mid-range systems.
- **Memory Limit:** ≤ 1 GB during runtime.
- **API Efficiency:** NLP API timeout fallback after 3s.
- **Decision Handling:** Support multiple unique choices per session without lag.

3.2 Scalability

- **Modular Architecture:** New levels or modules can be added without major rewrites.
- **API Flexibility:** Allow switching between APIs.
- **Data Logging:** Support migration from local to cloud storage.
- **Expandable Content:** Base design supports multiple levels.

3.3 Usability

- **Simple Interface:** Text input and options navigable via keyboard.
- **Consistent Layout:** Unified UI templates for dialogues and journals.
- **Minimal Learning Curve:** Players begin interacting within a few minutes.
- **Clear Feedback:** Visual or textual cues for processing and sentiment classification.

3.4 Reliability & Availability

- **Auto-Save:** Triggered after major decisions or transitions.
- **Crash Recovery:** Resume from last checkpoint after failure.

3.5 Security

- **Secure API Calls:** HTTPS + token authentication.
- **No Personal Data Collection:** Anonymous analytics only.
- **Anonymized NLP Input:** Strip identifiers before API calls.

3.6 Portability & Compatibility

- **Supported OS:** iOS/Android
- **Cross-Platform Build:** windows 10/11 for the future
- **Library Independence:** Use replaceable or platform-agnostic NLP dependencies.

3.7 Maintainability & Extensibility

- **Modular Components:** Sentiment module can be upgraded without altering core code.
- **Code Documentation:** Inline comments and Unity docs for maintainability.
- **Version Control:** Git-based collaboration with feature branching.
- **Defined Module APIs:** Standardized data exchange format between systems.

3.8 Testing & Quality Assurance

- **Unit Testing:** Verify sentiment → branch mapping and game logic.
- **User Testing:** Surveys for engagement and reflection quality.
- **Playtesting:** Minimum 10 participant sessions.
- **Regression Testing:** Ensure prior levels remain consistent post-updates.

3.9 Data Integrity & Backup

- **Save File Integrity:** Validate structure and variables on load.
- **Automatic Backups:** Triggered before level transitions.
- **Data Validation:** Detect and recover from corrupted saves.
- **Analytics Backup:** Export logs to CSV/cloud.
- **Data Retention:** Keep anonymized research data for 6 months.
- **Transaction-Based Saves:** Prevent partial writes during crashes.